

HealthParse: A Modular Pipeline for Privacy-Aware Extraction of Structured Clinical Information from Heterogeneous Healthcare Documents

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Abstract—A hospital intake desk can receive a cleaned typeset lab PDF, a faxed discharge summary, and a phone photo of a handwritten prescription in the same hour, and all must become useful clinical data rather than just be stored as incomprehensible files. HealthParse is a document processing pipeline designed to tackle such mixed document types, carrying a healthcare document through four stages: ingestion and validation, format-aware text extraction (native-text PDFs through direct parsing, scanned or photographed documents through Tesseract OCR and image preprocessing), privacy-aware de-identification of the recovered text, and hybrid clinical information extraction (a zero-shot named-entity model plus deterministic rules for values like dosages and lab results). Instead of showing only the conceptual architecture of this flow, this paper reveals what each stage actually produces (the data structures, confidence scores, and character-level provenance it retains) and reports an end-to-end evaluation on a six-document corpus mixing native-text PDFs and scanned images: all six documents were processed successfully, the pipeline produced 164 structured clinical entities across eight categories, and the privacy layer recorded 484 identifier redactions across eleven identifier types, at a mean end-to-end processing time of 5.83 seconds per document. These results are framed as operational execution results rather than an accuracy benchmark against labeled ground truth, and we are always explicit about that distinction. This paper is scoped deliberately to the ingestion-through-extraction chain and not to the full table-structure recognition or guaranteed removal of all identifying information or any kind of clinical decision support.

Keywords—Healthcare Document Processing, Optical Character Recognition, Clinical Named Entity Recognition, Medical Information Extraction, Privacy-Preserving De-identification, Document Intelligence

I. INTRODUCTION

Walk into most healthcare administrative offices and you will find records that don't share a common format [1], [2]. A lab returns a digitally generated PDF report containing neatly embedded text. A referring clinic faxes an attached discharge summary, scanned in as a low-quality, wavy image. A patient photographs a handwritten prescription on their phone, then uploads the picture through a portal. All three are essentially the same kind of information, but only the first is machine-parsable directly. The last two (and many similar ones) require an intermediary step of optical character recognition before any downstream system can do anything with them, which is far from a perfect process: its accuracy is

highly dependent on scan quality, lighting conditions, document skew, and other factors [3], [4].

This is a nontrivial limitation. Once a document is converted to text, it typically still contains identifying patient information, which must be handled appropriately before any of the content can be used for its intended purpose. And reliably performing all this processing, for documents in a variety of formats the system may not have been specifically trained on, is a significant engineering task in and of itself, distinct from and arguably more challenging than training an extraction model capable of doing so.

This paper describes the HealthParse document processing pipeline, designed to handle precisely this task. Given a healthcare-related document in one of several common formats, the pipeline ingests and validates the document, extracts any embedded machine-readable text (using native PDF tools when available) or performs OCR (with Tesseract [5], following an image preprocessing sequence [6], [7]) to obtain text from the document image, de-identifies the extracted text, then extracts structured clinical information from the resulting text using a combination of zero-shot named-entity recognition and deterministic pattern extraction (for example, for lab results or dosages).

This paper is scoped to focus specifically on this chain: ingestion and OCR/preprocessing, privacy-preserving de-identification of the extracted text, and extraction of the relevant clinical information. The pipeline has additional capabilities not described in this paper, most notably document classification, that fall outside the scope of this discussion, which we explicitly revisit in Section VIII. Within the scope of this paper, its contribution is:

- Description of the implemented pipeline at the level of concrete data structures/API responses, covering processing decisions that go beyond a conceptual architecture diagram
- Detailed discussion of the OCR/preprocessing stage's dual processing path for native-text PDFs vs scanned/image documents, covering what layout information gets preserved (rather than discarded) during text extraction
- Description of the privacy layer's capabilities and limitations: what sort of information it was designed to reduce exposure of, and an acknowledgment that automated de-identification of this sort can both over-redact and under-redact

- Description of the hybrid clinical extraction stage combining zero-shot NER with deterministic rules across eight clinical entity categories, including discussion of the character-offset provenance tracking that links each extracted entity to a location in the original text
- End-to-end system evaluation on a six-document test set containing a mix of native PDFs and scanned images, reporting overall execution success rate, per-stage timing breakdown, entity extraction counts by label, and privacy redactions applied, along with discussion of what lies outside this system’s scope of evaluation

II. RELATED WORK

A. OCR for Healthcare Documents

OCR has been studied for general digitization of documents [8]–[11], and for specific domains (invoices [12]–[14], historical archives [15], [16]), but relatively little work has focused on healthcare-specific benchmarking [17], [18]. The literature agrees that OCR performance is not a static value, but varies with the quality of documents (their layout complexity) and the degree of degradation in the image (the scan), and that errors at this stage can affect subsequent text processing steps [3]–[5]. We do not attach specific accuracy figures to this general statement, since the exact values reported in any one study are specific to that study’s data and evaluation setup and cannot be directly applied to a different document population.

B. Clinical Named Entity Recognition

The extraction of structured clinical entities from free text has traditionally been performed using supervised models trained on annotated corpora from the relevant clinical domain [19], [20]. Recent advances in zero-shot and few-shot span-identification models like GLiNER [21] enable the specification of a flexible label set at inference time without requiring retraining, at the cost of some raw accuracy in favor of being able to adapt without fresh labeled data to new entity categories. Deterministic, rule-based extraction remains common practice alongside learned models for structured value types —lab measurements, dosages, dates —where a pattern-based approach can often yield better results and be more interpretable than a learned model would be for that narrow category.

C. Privacy-Preserving De-identification

De-identification of clinical text is often approached as a mixture of rule-based recognizers for identifiers that take a predictable form (phone numbers, dates, structured ID numbers) and statistical or learned recognizers for identifiers that appear in unstructured free text, such as patient names [22]. Toolkits such as Microsoft Presidio [23] realize this mixture as a configurable pipeline of recognizers and anonymization operators and are commonly enhanced with custom recognizers for identifiers common to a given domain, such as medical record numbers in a healthcare setting. It is well understood in this literature that there is no automated de-identification approach that provides a guarantee of complete removal; false negatives (missed identifiers) and false positives (over-redaction of non-identifying text) are both known failure modes, rather than edge cases.

D. Positioning of This Work

Much of the existing literature in this space works on one stage (say, an OCR engine comparison) in isolation, with far less said about how these stages actually integrate into a single running pipeline and the implementation choices that this integration entails (how to represent OCR layout metadata once extraction moves to a downstream stage, how to pass de-identified text forward while retaining document structure, how to retain a traceable link from extracted entities back to their source). This paper’s contribution is at the level of integration, articulated concretely and demonstrated on a mixed-format document evaluation set rather than in the abstract.

III. SYSTEM ARCHITECTURE

HealthParse is implemented as a layered application with a frontend client submitting documents to a FastAPI backend that orchestrates downstream processes related to document ingestion, OCR/preprocessing, privacy de-identification, and clinical information extraction, with results persisted to a relational database along with the originally uploaded files. Fig. 1 provides an overview of the processing pipeline from the perspective of a single document processed by the application, from ingestion, through document-type-specific OCR (M2), through privacy-aware de-identification, to the structured clinical entities that the pipeline ultimately produces (M4). Sections IV and V discuss each of these stages in turn.

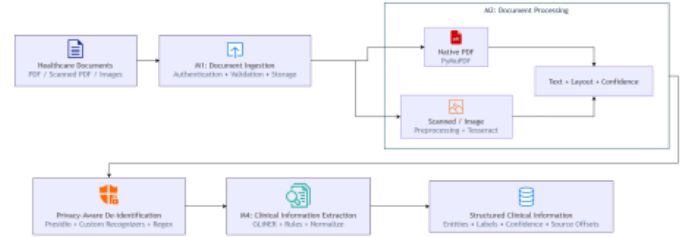


Fig. 1. End-to-end HealthParse pipeline: document ingestion (M1), document processing/OCR (M2), privacy-aware de-identification, and hybrid clinical information extraction (M4), producing structured clinical information with entity labels, confidence scores, and source offsets.

IV. METHODOLOGY

A. Document Ingestion (M1)

Documents are uploaded via a REST endpoint, which accepts PDF, JPEG, and PNG file types; each uploaded file is validated against the MIME type that was declared, and then a UUID is generated and associated with the uploading user (and an optional patient identifier), before the file is written to storage. A document record is created with a Processing status, and the remaining pipeline stages are scheduled in an asynchronous fashion, instead of being executed inline with the upload request. This design decouples document ingestion latency from the time required for OCR and subsequent stages to complete. We refer to this stage as ingestion and registration, rather than upload, because this behavior affects how the rest of the pipeline is composed and reasoned about.

B. Preprocessing and OCR (M2)

Once a document is queued for processing, the system decides between two extraction paths depending on document type. Digitally generated PDFs that already contain embedded text are handled directly through PyMuPDF, which extracts that text without invoking OCR at all —there is nothing to recognize visually when the text is already encoded in the file. Scanned PDFs and image uploads (JPEG, PNG) take a

different route: each page is rendered at 300 DPI and passed through an image-preprocessing sequence [6], [7] —grayscale conversion, coarse orientation correction, fine deskewing, denoising, and adaptive binarization —before Tesseract [5] performs the actual character recognition.

What happens after we recognize the text is just as important as the recognition itself. Instead of keeping a flat block of recognized text, the implementation preserves Tesseract’s per-word output: recognized text, confidence, a bounding box, and page/block/paragraph/line/word indices for every word on the page. We then use these word coordinates to reconstruct the document’s final raw text, instead of taking it as-is from Tesseract’s default reading order. This approach attempts to do better than just concatenating recognition output from top to bottom, and has not been benchmarked against the default ordering on pages with multiple text regions. The layout structure will take the form:

```
{
  "pages": 1,
  "page_dimensions": [...],
  "page_confidences": [...],
  "words": [...]
}
```

with each word entry recorded as:

```
{
  "page": 1,
  "text": "Medical",
  "confidence": 0.87,
  "bbox": {
    "x": 120,
    "y": 185,
    "width": 72,
    "height": 21
  }
}
```

This full structure —raw text, per-document confidence, page count, the OCR engine identifier, and processing time —is persisted alongside the document and exposed through a dedicated text-retrieval endpoint, so that both the recognized text and its spatial provenance remain available to downstream stages rather than being discarded once extraction completes. One limitation is worth stating plainly here rather than leaving implicit: M2 does not perform table-structure recognition. Reading order across a page is reconstructed from word coordinates, but genuine table and cell structure —the kind found in dense laboratory report grids —is not recovered as a table; it is recovered as positioned text.

C. Privacy Processing

The text produced by M2, whether from native PDF parsing or OCR, is run through a de-identification layer before any downstream processing touches it (Fig. 2). This layer is based on Microsoft Presidio’s analyzer [23], augmented with custom recognizers for identifiers common to a healthcare setting that Presidio doesn’t have out of the box —patient name, medical record number (MRN), unique health identifier (UHID), patient ID, and insurance or policy numbers —used in conjunction with Presidio’s built-in recognizers for phone numbers, email addresses, physical addresses, dates of birth, IP addresses, and account numbers. Detected identifiers are replaced with typed placeholder tokens such as [PATIENT], [MRN], [DATE], [PHONE], and [EMAIL] rather than being deleted outright, preserving the surrounding sentence structure for the extraction stage which comes next.

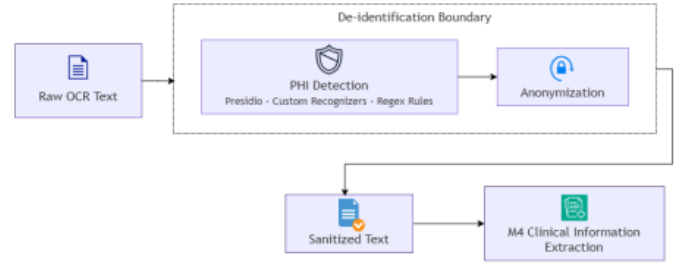


Fig. 2. Privacy de-identification boundary: PHI detection (Presidio, custom recognizers, and regex rules) followed by anonymization, producing sanitized text that is handed forward to M4 clinical information extraction.

We’re being careful about what we say about this stage. The privacy layer is intended to reduce exposure of identifiable information prior to downstream clinical processing; we make no claim here that all protected health information has been removed. Automated de-identification of this sort is known to produce false negatives —identifiers that go undetected —and false positives —non-identifying text that gets redacted unnecessarily —in general, and this implementation is no exception. Section VI-C and VI-D report what this layer actually did on our six-document evaluation set, as an operational redaction count rather than a measured precision/recall figure against a labeled identifier ground truth (which has as yet not been run).

D. Medical Information Extraction (M4)

The extraction stage consumes the sanitized output of the privacy layer and produces the pipeline’s clinical value (Fig. 3). Instead of using a single model, M4 uses two complementary extraction approaches. First, a zero-shot clinical NER model (GLiNER [21]) —urchade/gliner_small-v2.1 checkpoint in the current configuration —is used to detect mentions of eight clinical categories: Disease, Medication, Dosage, Symptom, Procedure, Anatomical Structure, Lab Test, and Lab Value with a detection threshold of 0.30. Second, deterministic rule-based extraction is used to extract value types that are more likely to have consistent patterns than the free text —laboratory values, dosages in context, and named laboratory tests —where pattern matching is likely to be more accurate and easier to audit than a learned model’s output for the same category.

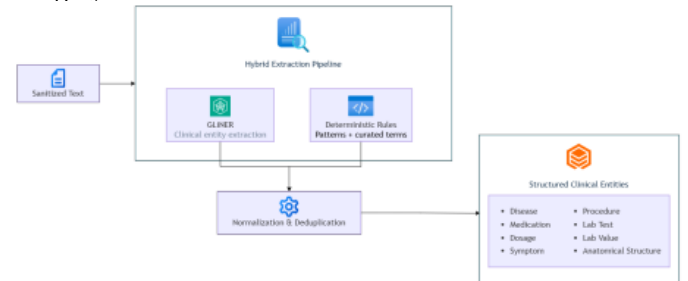


Fig. 3. M4 hybrid extraction pipeline: sanitized text is processed in parallel by GLiNER (zero-shot clinical entity extraction) and a deterministic rule engine, with results merged through normalization and deduplication into structured clinical entities across eight categories.

Results from both mechanisms are merged by a normalization step and a deduplication step before being stored as final structured entities. Each such stored entity retains its extracted text span, its label, a confidence score, and character start/end offsets into the sanitized source text (not merely the extracted value). It is these character offsets that

would enable an entity such as an extracted medication name to be traced back to the specific position in the text from which it was extracted, which in turn (in conjunction with the OCR bounding-box information retained in M2) would allow a downstream stage to resolve that entity into a specific region of the original document image. We are careful to note only what is implemented at this stage: extracted entities retain character offsets into the sanitized text, and M2 retains the spatial OCR metadata necessary to enable source-region resolution, but full resolution back to a highlighted document region is not evaluated in this paper.

V. IMPLEMENTATION

The pipeline outlined above has been implemented as a FastAPI backend in Python. Native-PDF text extraction is performed using PyMuPDF, while for scanned documents and images, Tesseract OCR is used with an accompanying OpenCV-based preprocessing chain that includes grayscale conversion, orientation detection and correction, deskewing, denoising, and adaptive binarization. The privacy de-identification component is based on Microsoft Presidio’s analyzer and anonymizer with custom healthcare-specific recognizers added on top. For medical information extraction, we use GLiNER (urchade/gliner_small-v2.1) in conjunction with a custom deterministic rule-based system for lab values, dosages, and lab tests. Extracted text, OCR layout, and entities are all stored in a relational database, with layout and entities being stored as JSON/JSONB, and exposed through a set of versioned REST endpoints for uploading documents, retrieving text, and retrieving entities.

VI. SYSTEM DEMONSTRATION AND EVALUATION

Consistent with this paper’s scope, we evaluate the implemented M1—M4 pipeline’s actual execution behavior on a fixed six-document set, rather than presenting a model-performance benchmark against labeled ground truth as the central result. The evaluation set intentionally mixes format types: two native-text PDFs (a 13-page and a 19-page clinical report) and four scanned/image documents (three JPEGs and one scanned PDF) processed through the Tesseract OCR path. This section covers functional verification, aggregate pipeline execution results, OCR/preprocessing characteristics, privacy de-identification results, clinical entity extraction results, and a single worked qualitative example. Where a quantity was not captured by this evaluation run, we say so directly in Section VI-G rather than filling in a plausible-sounding number.

Evaluation Environment: The evaluation was conducted on a 12th Gen Intel Core i5-1235U CPU with 8 GB RAM and integrated Intel UHD Graphics (CPU-only configuration; no discrete GPU was used). The software environment used Python 3.10.0, Tesseract OCR 5.5.3.20260724 (Leptonica 1.87.0), and the GLiNER model urchade/gliner_small-v2.1. The same environment and processing configuration were used for all six documents.

A. Functional Verification

Before reporting on pipeline output, we verify correct end-to-end operation of each stage. The implementation’s test suite exercises document ingestion and validation, the native-PDF and OCR extraction pathways, privacy de-identification, and the extraction and merging logic in M4, and is executed as part of the project’s standard verification

procedure ahead of any evaluation results reported in this section. We view functional verification —confirmation that each stage operates as designed on its test inputs —as a separate consideration from the output-quality results that follow.

B. Aggregate Pipeline Execution Results

Using the six document evaluation set, we processed all six documents successfully with no pipeline failures, logging zero rows in the error record, for 100% execution success rate for this set. Table I shows the complete per-document breakdown, including input format and page count, the OCR/extraction engine used and its confidence, per-stage processing time (M2 OCR/preprocessing, privacy de-identification, and M4 extraction), total processing time, and the number of structured entities produced.

TABLE I
PER-DOCUMENT EXECUTION RESULTS ACROSS THE SIX-DOCUMENT
EVALUATION SET

Document (type)	Pg.	Engine	M2 (ms)	Priv. (ms)	Tot. (ms)	Ent.
PK0016 (PDF)	13	native	143	2059	12503	68
Sterling path. rpt. (PDF)	19	native	204	3135	4048	33
Tmpl. 1 sample 1 (PDF)	1	Tesseract	6126	218	6985	15
Tmpl. 2 sample 5 (JPG)	1	Tesseract	3343	167	4092	10
Tmpl. 4 sample 1 (JPG)	1	Tesseract	3375	201	4161	23
Tmpl. 7 sample 7 (JPG)	1	Tesseract	2750	113	3182	15
Mean	—	—	2657	982	5829	27.3
Median	—	—	—	—	4126	—

All six documents were marked SUCCESS, with a mean total processing time of 5828.55 ms and a median of 4126.48 ms; the fastest document completed in 3182.28 ms and the slowest —the 13-page native PDF, driven by its M4 extraction time of 10301.76 ms against a much larger volume of embedded lab data —completed in 12502.96 ms. Averaged across the set, M2 (preprocessing/OCR) accounted for a mean of 2656.87 ms, privacy processing for 982.18 ms, and M4 extraction for 2189.48 ms per document. We report these as an operational demonstration of the implemented pipeline’s execution characteristics on this six-document set, not as a claim about throughput on an arbitrary document population.

C. OCR and Preprocessing Characteristics

The two native-text PDFs were parsed with a reported PyMuPDF confidence of 1.0000, since no visual recognition was required. The four scanned/image documents processed through Tesseract reported per-document confidence between 0.9301 and 0.9478, consistent across formats despite differing image quality and content density. As emphasized in Section IV-B, this confidence score reflects Tesseract’s own certainty about its reading, not a measured accuracy against ground truth; it is a signal for flagging text worth closer review rather than a substitute for one. The preprocessing sequence —grayscale conversion through adaptive binarization —is applied uniformly ahead of Tesseract on the scanned/image path, and its qualitative effect is to convert a photographed or scanned page into the more uniform, higher-contrast input that Tesseract performs best on.

D. Privacy De-identification Results

Across the six documents, the privacy layer recorded 484 total redactions spanning eleven identifier categories:

DATE_TIME, PERSON, PATIENT_NAME, PATIENT_ID, ACCOUNT_NUMBER, LOCATION, IP_NUMBER, PHONE_NUMBER, ADDRESS, DATE_OF_BIRTH, and MRN. Redaction volume scaled with document length and identifier density rather than format: the 19-page pathology report produced the most redactions (247, dominated by 98 PERSON and 107 DATE_TIME detections), while the single-page scanned documents each produced between 21 and 34 redactions. As in Section IV-C, we report this as the layer’s recorded operational output —what it detected and replaced —rather than as a measured precision or recall figure, since no labeled identifier ground truth evaluation has been run against this set.

E. Clinical Information Extraction Results

M4 identified 164 structured clinical entities in the six documents, populating all eight supported entity labels. Table II shows the distribution; Fig. 4 shows the same distribution graphically. Lab value (56) and lab test (51) entities dominated the two native-PDF laboratory reports, where deterministic pattern rules are most applicable to the structured numeric+unit format those documents use (e.g., “192 mg/dL”, “eGFR”). Symptom, anatomical structure, disease, and procedure entities —the categories that most rely on the zero-shot GLiNER model rather than pattern rules —were concentrated in the four scanned clinical-note and radiology-style documents, where clinical findings are expressed in free text rather than tabular values.

TABLE II
EXTRACTED ENTITY COUNTS BY LABEL ACROSS THE SIX-DOCUMENT SET

Entity label	Count	Docs. present in
Lab Value	56	4/6
Lab Test	51	3/6
Symptom	18	4/6
Anatomical Structure	16	4/6
Procedure	14	6/6
Disease	5	2/6
Medication	3	3/6
Dosage	1	1/6
Total	164	—

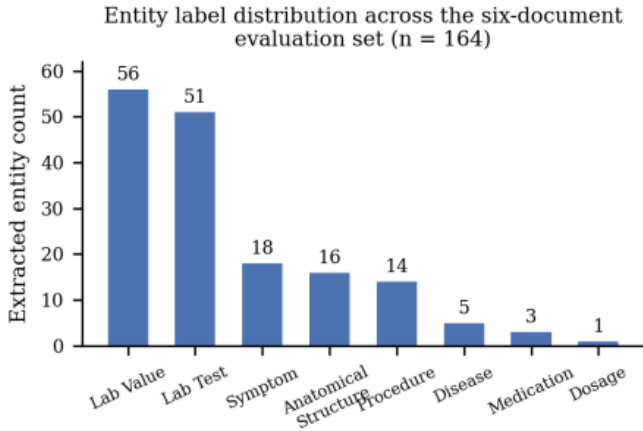


Fig. 4. Distribution of the 164 structured clinical entities extracted across the six-document evaluation set, by entity label.

Procedure was the only label which appeared in all six documents, used in contexts ranging from a microscopy laboratory procedure (“Microscopy”) to surgical and

diagnostic procedures (“Wide local excision”, “Slit-lamp Examination”, “Tonometry”), reflecting that some description of a procedure performed is common to lab reports, dermatology/pathology notes and clinical encounter notes alike. Every entity record retained a label, a confidence score, and character start/end offsets into the sanitized source text, consistent with the provenance design described in Section IV-D; individual entity confidences in this run ranged from as low as 0.30 (the GLiNER detection threshold) for ambiguous free-text spans up to 0.95 for deterministically matched lab values.

F. Qualitative End-to-End Example

The following is an illustrative system-output example constructed to show the pipeline’s stage-to-stage behavior on a single sentence; it is not one of the six measured evaluation records reported in Sections VI-B through VI-E. Consider a representative sentence of the kind M2 recovers from a scanned document: “Patient John Doe, DOB 04/12/1985, MRN 88213, was prescribed Metformin 500 mg for diabetes.” The privacy layer (Section IV-C) reduces this to “Patient [PATIENT], DOB [DATE], MRN [MRN], was prescribed Metformin 500 mg for diabetes,” leaving the clinical content untouched. M4 then extracts three structured entities from that sanitized sentence:

```
{ "text": "Metformin", "label": "MEDICATION",
  "start_char": 61, "end_char": 70 }
{ "text": "500 mg", "label": "DOSAGE",
  "start_char": 71, "end_char": 77 }
{ "text": "diabetes", "label": "DISEASE",
  "start_char": 82, "end_char": 90 }
```

Each of these three entities is present in, and only in, the sanitized sentence —no clinical value appears in the extraction output that was not actually in the input text. GLiNER and the deterministic rule set jointly cover this example —MEDICATION and DISEASE come from the zero-shot model, while DOSAGE is well suited to the pattern-based rule path —which is representative of how the two mechanisms divide labor across the eight supported categories in the six-document set as a whole (Section VI-E).

G. What This Evaluation Does Not Establish

We list explicitly what this six-document run does not establish, in keeping with the evidence-based approach this paper tries to hold itself to throughout. It does not provide manually annotated ground truth for entity-level precision, recall, or F1 on these documents; the entity and redaction counts reported above are the pipeline’s own operational output, not a measurement against an independent reference. It does not provide an independently validated privacy recall/precision figure; the 484 redactions reported in Section VI-D are recorded detections and anonymization operations, not a confirmed true-positive count. Successful execution on six documents should not be read as evidence of large-scale production scalability or of clinical reliability, and the set itself —two long native-PDF lab reports and four short scanned clinical notes —is small and was not constructed as a statistically representative sample of the document population a deployed system would encounter. A GLiNER-based clinical-entity extraction experiment and a related OCR-degradation analysis were carried out earlier in this project as separate internal evaluation artifacts; these experiments were conducted separately from the present system evaluation and are outside the scope of this paper.

VII. DISCUSSION

The results in Section VI demonstrate that heterogeneous healthcare documents —long native-text PDFs and short scanned images alike —can be processed through a single pipeline into structured, de-identified, and traceable clinical output, with the pipeline demonstrating zero recorded failures in processing all six evaluation documents. What makes the pipeline interesting as a system is not any given stage’s peak accuracy, but rather that the stages compose well with one another, and that this composability is reflected in the results: OCR output has retained the layout metadata that downstream traceability requires, privacy processing has performed 484 redactions while retaining 164 clinical entities within the same sanitized text for downstream extraction, and extraction has retained character offsets to enable auditable rather than opaque output. The split between entity categories and document types —lab values and lab tests concentrated within the two long structured PDF reports, and symptoms and anatomical findings concentrated within the four shorter free-text scans —is indicative of the division between deterministic extraction of structured clinical patterns and model-based extraction of free-text clinical concepts intended by the authors, though this finding constitutes a descriptive rather than controlled observation on the behavior of the two mechanisms.

VIII. LIMITATIONS AND FUTURE WORK

The scope of this paper is limited to the M1 through M4 portion of HealthParse —ingestion, OCR and preprocessing, privacy de-identification, and clinical information extraction. The broader project encompasses additional downstream capabilities including document-type classification that are not evaluated, diagrammed, or asserted to be a contribution of this paper; we mention this explicitly so it is not assumed to be in the scope of this paper. Extending from character-offset traceability to full resolution of an entity back to a highlighted region of the original document image is another concrete next step that builds on the OCR layout metadata that has already been preserved in M2. The current M4 configuration does not involve explicit long-document chunking. As a result, documents that exceed the model’s sequence length may be truncated during GLiNER inference while deterministic rules continue to apply to the entire sanitized text.

IX. CONCLUSION

This paper described HealthParse’s ingestion-through-extraction pipeline: document validation and asynchronous processing, dual native-PDF and OCR text extraction with layout fidelity and confidence metadata, Presidio-based privacy de-identification, and hybrid GLiNER-plus-rules clinical entity extraction with character-offset provenance. In addition to the implementation description, we reported an end-to-end evaluation on a six-document mixed-format corpus: 100% observed execution completion on the six-document evaluation set, mean 5.83 seconds of processing time per document, 164 structured clinical entities extracted across eight categories, and 484 recorded privacy redactions across eleven identifier types. The contribution here is the integrated, implemented pipeline and the concrete execution evidence collected against it —not a claim of state-of-the-art accuracy in any single stage, and not a claim that identifying information is guaranteed to be

removed. Future work includes a labeled-ground-truth accuracy evaluation for both de-identification and clinical entity extraction, expansion of the evaluation set beyond the present six documents, extending traceability to full source-region resolution, and, separately from this paper’s scope, the platform’s downstream document-classification capability.

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